VIDA JAMALI

vida@gatech.edu | +1 404-894-5134 | vidajamali.com

PROFESSIONAL APPOINTMENTS

Georgia Institute of Technology, Atlanta, GA

Assistant Professor, School of Chemical and Biomolecular Engineering Aug. 2022–present

Dr. James Robert and Margaret Spencer Early Career Fellow Apr. 2026–present

Daniel B. Mowrey Faculty Fellow Aug. 2022–present

University of California, Berkeley, Berkeley, CA

Postdoctoral Researcher ,

Department of Chemistry, Kavli Energy NanoScience Institute Dec. 2017–Aug. 2022

Advisor: A. Paul Alivisatos; Collaborator: Kranthi K. Mandadapu

Interdisciplinary Appointments and Affiliations

- Affiliated Faculty, Institute for Matter and Systems, Georgia Tech
- Affiliated Faculty, Institute for Data Engineering and Science, Georgia Tech
- Affiliated Faculty, Institute for Bioengineering and Biosciences, Georgia Tech
- Program Faculty, Machine Learning Ph.D. Program, Georgia Tech
- Program Faculty, Bioengineering Ph.D. Program, Georgia Tech
- Member, Strategic Energy Institute, Georgia Tech

EDUCATION

Rice University, Houston, TX

Ph.D., Chemical and Biomolecular Engineering 2017

Advisor: Matteo Pasquali

Thesis: *Morphology of Carbon Nanotube Liquid Crystalline Solutions: Insights into Tactoids and Columnar Phase*; Committee: Paul van der Schoot, S. Lisa Biswal, Fred C. MacKintosh

Sharif University of Technology, Tehran, Iran

B.Sc., Chemical Engineering 2011

HONORS AND AWARDS

Research Honors, Fellowships, and Professional Recognition

Digital Discovery Emerging Investigator	2026
Beckman Young Investigator Award finalist	2025
Scialog Fellow, Automating Chemical Laboratories, Research Corporation for Science Advancement	2024–2026
National Science Foundation CAREER Award	2024
ACS Petroleum Research Fund Doctoral New Investigator Award	2023
AI for Chemicals and Materials Discovery Initiative Lead, Georgia Tech	2023
35 Voices of Chemistry of Materials, American Chemical Society	2023
Rising Stars in Soft and Biological Matter, University of Chicago MRSEC	2021
Berkeley Postdoctoral Association Professional Development Award	2021

AIChE Women in Chemical Engineering Travel Award 2020
Future Faculty in Soft Matter Workshop, Princeton University and University of Delaware 2019
ACS P2F Future Faculty Scholar 2019
Society of Rheology Student Travel Award 2017
Active and Smart Matter Conference Travel Award 2016
Society of Iranian-American Women for Education Scholarship 2016
Smalley-Curl Institute Travel Award, Rice University 2016
ConocoPhillips Endowed Scholarship 2014
Phi Lambda Upsilon Honor Society 2014
NASA Space Health Innovation Challenge hackathon finalist 2013
Ignite Silicon Valley Trek Travel Award, Rice Alliance for Technology and Entrepreneurship 2013
Best Teaching Assistant Award, Department of Chemical and Biomolecular Engineering, Rice University 2012
Screech Elevator Pitch Competition People's Choice Award, Rice Center for Engineering Leadership 2012

PUBLICATIONS

* denotes equal contribution; † denotes corresponding author.

28. Panicker, I. A., Requejo, K. I., Ruiz, R., Ben-Moshe, A., Bustamante, C., Alivisatos A. P., Jamali, V.†
"Polarity-resolved imaging reveals active transport in catalytic Janus nanomotors."
ChemRxiv. (under review at *PNAS*) 2026

27. Patel, R., Briceno, G., Aghazadeh, A., Jamali, V.†
"Multimodal LLM-assisted diagnosis of cryptic failure modes in high-throughput experimentation: A case study in gold nanorod synthesis."
Neural Information Processing Systems, AI for Accelerated Materials Design Workshop 2026

26. Shabeeb, Z.*, Sreeram, S.*, Panicker, I. A., Wang, A., Xu, M., Goel, R., Jamali, V.†
"SAM-EM: Segment Anything for Liquid Phase Electron Microscopy Video Segmentation."
Digital Discovery, 5(9), 3625–3635 , Emerging Investigators Special Collection. Website: https://jamalilab.github.io/SAM-EM/ 2026

25. Shabeeb, Z.*, Saeedi, D.*, Tsui, D.*, Jamali, V.†, Aghazadeh, A.†
"Generative Priors for Cryo-EM Image Reconstruction."
ICML Workshop on Generative and Agentic AI for Biology 2026

24. Jamali, V.,†, Aghazadeh, A.†, Kacher, J.
"Thinking Microscopes: Agentic AI and the Future of Electron Microscopy."
npj Computational Materials 12, 149 2026

23. Panicker I.A., Shabeeb, Z., Hargus, C., Jamali, V.†
"Solution-Tunable Interfacial Interaction Landscape Governs Anomalous Nanoparticle Diffusion in Liquid-Phase Electron Microscopy."
ACS Nano 20, 21, 15550–15564 (Featured on front cover) 2026

22. Requejo, K. I., Cañari-Chumpitaz, C., Jamali, V., Ruiz, R., Alivisatos, A. P., Bustamante, C.
"Quantifying photochemical propulsion in light-powered Janus micromotors."
ACS Nano 20, 19, 14099–14112 2026

21. Shabeeb, Z.*, Saeedi, D.*, Tsui, D.*, Jamali, V.†, Aghazadeh, A.†
"cryoSENSE: Compressive Sensing Enables High-throughput Microscopy with Sparse and Generative Priors on the Protein Cryo-EM Image Manifold."
IEEE/CVF, Computer Vision and Pattern Recognition (CVPR). Website: https://cryosense.github.io 2026

20. Housley S., N.†, Bourque, A.R., Matyunina, L.V., Panicker, I.A., Schrader Echeverri, E., Izykowicz, M., Herrmann, O.A., Lee, S., Arboleda, J.C., Jamali, V., McDonald, J.F., Dahlman, J. E., Finn, M.G.
"Tumor Agnostic Drug Delivery with Dynamic Nanohydrogels."
Nature Communications, 17, 184 2026

19. Shabeeb, Z., Goyal, N., Attah Nantogmah, P., Jamali, V.†
"Learning the Diffusion of Nanoparticles in Liquid Phase TEM via a Physics-informed Generative AI."
Nature Communications 16, 6298 2025

18. Shabeeb, Z., Jamali, V.†
"LEONARDO: A Physics-Informed Generative Model for Stochastic Nanoparticle Dynamics in Liquid-Phase TEM."
Neural Information Processing Systems, AI for Science Workshop 2025

17. Wang*, A., Xu*, M., Goel, R., Shabeeb, Z., Panicker, I. A., Jamali, V.†
"SAM-EM: Real-Time Segmentation for Automated Liquid Phase Transmission Electron Microscopy."
Neural Information Processing Systems, AI for Accelerated Materials Design Workshop 2025

16. Shabeeb, Z., Goyal, N., Attah Nantogmah, P. Jamali, V.†
"Learning the Physics of Liquid Phase TEM Nanoparticle Trajectories Using Physics-Informed Generative AI."
Microscopy and Microanalysis 30, S1, 1724 2024

15. Shabeeb, Z., Goyal, N., Jamali, V.†
"Determining Diffusion Characteristics of Nanoparticles in Liquid Phase TEM Using Deep Learning."
Microscopy and Microanalysis 30, S1, 2074 2024

14. Jamali, V.†, Alivisatos, A. P.†
"Studying Diffusion of Colloidal Nanoparticles in Solution Using Liquid Phase TEM and Machine Learning."
Microscopy and Microanalysis 28, 142 - 143 2022

13. Moreno-Hernandez, I. A.*, Crook, M. F.,* Jamali, V.*, Alivisatos, A. P.
"Recent advances in the study of colloidal nanocrystals enabled by in situ liquid phase transmission electron microscopy."
MRS Bulletin 47 (3) 2022

12. Abbas, A., Vargo, E., Jamali, V., Ercius, P., Pieters, P., Brinn, R., Ben-Moshe, A., Cho, M., Xu, T., Alivisatos, A. P.
"Observation of an Orientational Glass in a Superlattice of Elliptically-Faceted CdSe Nanocrystals."
ACS Nano 16(6), 9339–9347 2022

11. Jamali, V., Hargus, C., Ben Moshe A., Aghazadeh, A., Ha, H. D., Mandadapu, K. K., Alivisatos, A. P.
"Deep Learning-assisted Liquid Cell Electron Microscopy Reveals the Nature of Anomalous Diffusion of Nanoparticles Near the Surface."
Proceedings of the National Academy of Sciences (PNAS) 118 (10) 2021

10. Jamali, V., Mirri, F., Biggers, E. G., Pinnick, R.A., Liberman, L., Cohen, Y., Talmon, Y., MacKintosh F., van der Schoot, P., Pasquali, M.
"Enhanced ordering in length-polydisperse carbon nanotube solutions at high concentrations as revealed by the small angle X-ray scattering."
Soft Matter 17, 5122-5130 *Featured on the front cover of Soft Matter, Issue 20* 2021

9. Cho, H., Moreno-Hernandez, I., Jamali, V., Oh, M., Alivisatos, A. P.
"In situ quantification of interactions between charged nanorods in a predefined potential energy landscape."
Nano Letters 21 (1), 628-633 2021

8. Jamali, V.*, Niroui, F.*, Taylor, L. W., Dewey, O. S., Koscher, B. A., Pasquali, M., Alivisatos, A. P.
"Perovskite-Carbon Nanotube Light Emitting Fibers."
Nano Letters 20 (5), 3178-3184 2020

7. Liberman, L., Jamali, V., Pasquali, M., Talmon, Y.
"The effect of carbon nanotube diameter and stiffness on their phase behavior in crowded solutions."
Langmuir 36 (1), 242-249 2020

6. Mirri, F.*, Ashkar, R.*, Jamali, V., Liberman, L., Pinnick, R., Talmon, Y., van der Schoot, P., Butler, P., Pasquali, M.
"Quantification of carbon nanotube liquid crystal morphology via neutron scattering."
Macromolecules 51 (17), 6892-6900 2018

5. Maillaud, L., Headrick, R. J., Jamali, V., Maillaud, J., Tsentalovich, D., Neri, W., Bengio, E. A., Mirri, F., Kleinerman, O., Talmon, Y., Poulin, P., and Pasquali, M.
"Flexible and conductive fibers made from highly concentrated aqueous dispersions of carbon nanotubes."
Industrial and Engineering Chemistry Research 57 (10), 3554-3560 2018

4. Tran, T. Q., Headrick, R. J., Bengio, E. A., Myint, S. M., Khoshnevis, H., Jamali, V., Duong, H. M., Pasquali, M.
"Purification and dissolution of carbon nanotube fibers spun from floating catalyst method."
ACS Applied Materials and Interfaces 9 (42), 37112-37119 2017

3. Jamali, V., Biggers, E., van der Schoot, P., Pasquali, M.
"Line tension of twist-free carbon nanotube lyotropic liquid crystal microdroplets on solid surfaces."
Langmuir 33 (36), 9115-9121 2017

2. Jiang, C., Peng, Z., de los Reyes, C., Young, C. C., Tsentalovich, D., Jamali, V., Ajayan, P. M., Tour, J. M., Pasquali, M., and Marti A. A.
"Increased solubility and fiber spinning of graphenide dispersions aided by crown-ethers."
Chemical Communications 53 (9), 1498-1501 2016

1. Jamali, V.*, Behabtu, N.*, Senyuk, B., Lee, J. A. Smalyukh, I.I., van der Schoot, P., Pasquali, M.
"Experimental realization of crossover in shape and director field of nematic tactoids."
Physical Review E 91 (4), 042507 2015

PATENTS AND INTELLECTUAL PROPERTY

3. Aghazadeh, A., Jamali, V., Saeedi, D., Shabeeb, Z., Tsui, D., "Compressive Sensing Enables High-throughput Microscopy with Sparse and Generative Priors on the Protein Cryo-EM Image Manifold", U.S. provisional application USSN 63/977,630

2. Alivisatos, A. P., Niroui, F., Jamali, V., Pasquali, M., "Light Emitting Fibers", U.S. provisional application USSN 62/714,561

1. Alivisatos, A. P., Jamali, V., "Processing Method for Fabricating Perovskite-Carbon Nanotube Fibers and Devices", U.S. provisional application USSN 62/958,394

RESEARCH SOFTWARE AND OPEN-SOURCE RESOURCES

cryoSENSE: Compressive sensing with sparse and generative priors on the protein cryo-EM image manifold. Project website

SAM-EM: Segment Anything for liquid-phase electron microscopy video segmentation. project website; Project

website

LEONARDO: Physics-informed generative modeling of nanoparticle diffusion in liquid-phase TEM. Project website

MotionNet(MoNet): Physics-informed deep learning for classifying single-particle trajectories and estimating the anomalous diffusion exponent. Project website

FUNDED RESEARCH GRANTS AND SUPPORT

Principal Investigator

Research Corporation for Science Advancement,*Closing the Characterization Loop in Autonomous Materials Discovery: Agentic AI for Multimodal Instrument Orchestration* 2026–2027
PI; direct funding $66,000; Aug. 1, 2026–July 30, 2027

Foresight Institute,*Self-Driving Nanofabrication with Thinking Microscopes* 2026–2027
PI; direct funding $10,000; Oct. 1, 2026–Sept. 30, 2027

Microsoft Azure,*Experiment-in-the-Loop Agentic AI for Task-Adaptive Self-Driving Labs* 2026–2027
PI; $12,500 in computing support; Oct. 1, 2026–Sept. 30, 2027

National Science Foundation (NSF CAREER),*CAREER: Imaging and Understanding the Motion and Interaction of Nanoparticles Near Surfaces* 2024–2028
PI; total award $744,338; Jan. 1, 2024–Dec. 30, 2028

Georgia Tech Institute for Electronics and Nanotechnology,*Compressed Super-Resolution TEM Using a Nanoelectronic Coded Aperture Device* 2024–2026
PI; co-PI: Amirali Aghazadeh; total award $225,000 in direct costs; Jan. 1, 2024–Dec. 30, 2026

American Chemical Society Petroleum Research Fund (DNI),*Investigating Rotational and Translational Diffusion of Nanoparticles at Liquid–Liquid Interfaces via Time-Resolved Nanoscale Imaging* 2023–2025
PI; total award $110,000; Sept. 1, 2023–Aug. 31, 2025

Co-Principal Investigator

Georgia Tech Institute for Data Engineering and Science and Institute for Materials, *AI for Chemicals and Materials Discovery* 2024
Co-PI; collaborators: Victor Fung, Pan Li, Amirali Aghazadeh; total award $20,000 Jan. 1–Dec. 1, 2024

INVITED TALKS AND RESEARCH PRESENTATIONS

Upcoming Invited Talks

LPTEM Reveals Active Transport and Polarity-Dependent Propulsion of Catalytic Janus Nanomotors *(invited)* Apr 2027
MRS Spring Meeting, Seattle, WA

Invited Conference, Symposium, and Workshop Talks

AI-Driven Electron Microscopy: Imaging, Learning, and Understanding Nanoscale Dynamics *(invited)* Sep 2026
Venice meeting on fluctuations in small complex systems VIII, Venice, Italy

AI-Assisted LPTEM: A Unified Framework for Nanoparticle Tracking and Generative Modeling *(invited)* Jan 2026
Gordon Research Conference on Liquid Phase EM, Ventura, CA

Liquid phase transmission electron microscopy for investigating multiscale dynamics of biomolecular systems Aug 2025
1st SouthEast Amyloid-CONdensate (SEACON) symposium, Ocean Springs, MS

On the Viscoelastic Origin of Nanoparticle Surface Diffusion in LPTEM *(invited)* May 2025
Southeast Soft Matter Day, Atlanta, GA

Imaging, Learning, and Engineering of Nanoscale Dynamics via LPTEM *(invited)* Apr 2025
MRS Spring Meeting, Seattle, WA

Using Deep Learning to Elucidate the Dynamics of Nanoparticles in LPTEM *(invited)* Sep 2023
International Microscopy Congress IMC20, Busan, Korea

Deep Learning-Assisted Analysis of Anomalous Nanoparticle Surface Diffusion in LPTEM *(invited)* Mar 2023
2023 Institute for Materials Symposium on Materials Innovations, Atlanta, GA

Deep Learning-Assisted Analysis of Anomalous Nanoparticle Surface Diffusion in LPTEM *(invited)* Mar 2023
CCMST, Machine Learning in Chemistry Seminar, GeorgiaTech, Atlanta, GA

Deep Learning-Assisted Analysis of Anomalous Nanoparticle Surface Diffusion in LPTEM *(invited)* Oct 2022
LPTEM Gordon Research Conference, Ventura, CA

Studying Diffusion of Colloidal Nanoparticles in Solution Using LPTEM and Machine Learning *(invited)* Aug 2022
Microscopy and Microanalysis Conference, Portland, OR

Imaging, learning, and engineering of complex soft matter systems at the nanoscale Sep 2021
Rising Stars in Soft and Biological Matter Symposium, University of Chicago MRSEC (virtual)

Morphology of carbon nanotube liquid crystalline phases: insight into tactoids and columnar phase *(invited)* Mar 2018
APS March Meeting, Los Angeles, CA

Invited Departmental Seminars, Institutional Lectures, and Webinars

AI-Driven Electron Microscopy for Visualizing Nanoscale Dynamics in Liquid *(invited)* May 2026
NSF Center for Chemical Innovation (CCI), University of Chicago, Chicago, IL (virtual)

Decoding Nanoscale Motion: AI-Driven Liquid Phase TEM *(invited)* Jan 2026
Protochips Webinar on Exploring Synthesis of Nanomaterials Using Liquid Phase TEM (Virtual)

AI-Driven Electron Microscopy for Visualizing Nanoscale Dynamics in Liquid *(invited)* Dec 2025
University of California, Santa Barbara, CA

AI for Anomalous Diffusion: From Classification to Generative Modeling of Stochastic Transport at the Nanoscale Oct 2025
Potsdam University, Potsdam, Germany (Invited, Virtual)

Transforming electron microscopes to automated factories of discovery May 2025
Materials x Robotics Initiative, Institute for Matter and Systems, Georgia Tech, Atlanta, GA

Learning the Diffusion of Nanoparticles in LPTEM Using Physics-Informed Generative AI *(invited)* Nov 2024
Ohio State University-Protochips Webinar on Liquid Cell TEM, Columbus, OH

Imaging, Learning, and Engineering of Nanoparticles' Dynamics Using LPTEM *(invited)* Mar 2024
Institute for Electronics and Nanotechnology, Georgia Tech, Atlanta, GA

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Mar 2022
University of Southern California, Chemical Engineering and Materials Science, Los Angeles, CA

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Feb 2022
Massachusetts Institute of Technology, Department of Chemical Engineering, Cambridge, MA

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Feb 2022
Brandeis University, MRSEC, Waltham, MA

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Feb 2022
University of Minnesota, Department of Chemical Engineering and Materials Science, Minneapolis, MN

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Feb 2022
University of Wisconsin-Madison, Department of Chemistry, Madison, WI

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Feb 2022
University of California, Los Angeles, Department of Chemistry and Biochemistry, Los Angeles, CA

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Feb 2022
Yale University, Department of Chemical and Environmental Engineering, New Haven, CT

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Jan 2022
Cornell University, Department of Chemistry and Chemical Biology, Ithaca, NY

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Jan 2022
Princeton University, Department of Chemical and Biological Engineering, Princeton, NJ

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Jan 2022
Georgia Institute of Technology, School of Chemical and Biomolecular Engineering, Atlanta, GA

Imaging, learning, and engineering of soft matter systems at the nanoscale *(invited)* Dec 2021
University of California, Berkeley, Kavli Energy NanoScience Institute, Berkeley, CA

Deep learning-assisted analysis of anomalous nanoparticle surface diffusion in liquid phase TEM *(invited)* Sep 2021
University of California, Berkeley, Nano Seminar Series, Berkeley, CA

Imaging, learning, and engineering of complex soft matter systems at the nanoscale *(invited)* Jul 2021
Seagate Normandale AI/ML Distinguished Seminar Series, Bloomington, MN (virtual)

Imaging, learning, and engineering of complex soft matter systems at the nanoscale Feb 2021
Department of Chemical and Biomolecular Engineering, Johns Hopkins University, MD (virtual)

Imaging, learning, and engineering of complex soft matter systems at the nanoscale Feb 2021
Department of Chemical Engineering, Texas A&M University, TX (virtual)

Imaging, learning, and engineering of complex soft matter systems at the nanoscale Jan 2021
Department of Chemical and Biomolecular Engineering, University of Notre Dame, IN (virtual)

From carbon nanotube liquid crystalline solutions to functional fibers *(invited)* Feb 2019
Department of Materials Science and Engineering, Cornell University, Ithaca, NY

From carbon nanotube liquid crystalline solutions to functional fibers *(invited)* Nov 2018
Department of Physics, Physics of Living Systems, MIT, Cambridge, MA

Phase behavior and morphology of carbon nanotube liquid crystal solutions *(invited)* Feb 2017
Lewis-Sigler Integrative Genome Institute, Biophysics group, Princeton, NJ

Other Conference and Workshop Presentations

SAM-EM: Real-Time Segmentation for Automated Liquid Phase Transmission Electron Microscopy [Spotlight] Dec 2025
AI4Mat workshop, Neural Information Processing Systems, San Diego, CA

AI-enabled Single Particle Tracking in Liquid-Phase Transmission Electron Microscopy Jun 2025
ACS Colloid & Surface Science Symposium, Alberta, Canada

Physics-informed Generative Models for Learning the Stochastic Diffusion of Single Particle Trajectories Mar 2025
American Physical Society March Meeting, Anaheim, CA

Liquid Phase Transmission Electron Microscopy for Single Particle Tracking Oct 2024
AIChE Fall Meeting, San Diego, CA

Deep Generative Neural Networks for Learning the Dynamics of Nanoparticles in LPTEM Mar 2024
APS March Meeting, Minneapolis, MN

Deep Learning-Assisted Analysis of Anomalous Nanoparticle Surface Diffusion in Liquid Phase TEM Nov 2021
AIChE Annual Meeting, Boston, MA

Deep learning-assisted analysis of anomalous nanoparticle diffusion near the liquid cell surface reveals the effect of electron beam dose rate in TEM Nov 2021
AIChE Annual Meeting, Boston, MA

Deep learning-assisted analysis of anomalous nanoparticle surface diffusion in liquid phase TEM Jun 2021
ACS Colloid & Surface Science Symposium (virtual)

Deep learning-assisted analysis of anomalous nanoparticle diffusion near the liquid cell surface reveals the effect of electron beam dose rate in TEM Mar 2021
APS March Meeting (virtual)

In-situ liquid phase electron microscopy for studying the dynamics of colloidal nanoparticles at the nanoscale Nov 2020
AIChE Annual Meeting, San Francisco, CA (virtual, available online)

From nanoscale building blocks to functional fibers Nov 2019
AIChE Annual Meeting, Orlando, FL

Colloidally synthesized nanomaterials as building blocks for functional fibers Nov 2018
MRS Fall Meeting, Boston, MA

From carbon nanotube liquid crystalline solutions to functional fibers Oct 2018
AIChE Annual Meeting, Pittsburgh, PA

A hexagonal columnar liquid crystal phase formation in dilute solutions of carbon nanotubes Oct 2017
AIChE Annual Meeting, Minneapolis, MN

Phase behavior and morphology of carbon nanotube liquid crystal solutions Feb 2017
88th Society of Rheology Conference, Tampa, FL

Morphology of carbon nanotube liquid crystal solutions Nov 2016
AIChE Annual Meeting, San Francisco, CA

Wetting behavior, shape, and morphology of sessile lyotropic liquid crystal microdroplets Jun 2016
ACS Colloid & Surface Science Symposium, Harvard University, Cambridge, MA

Wetting behavior, shape, and morphology of sessile lyotropic liquid crystal microdroplets Apr 2016
SCI Transdisciplinary Symposium, Rice University, Houston, TX

Experimental realization of crossover in shape and director field of nematic tactoids Aug 2015
Texas Soft Matter Meeting, Houston, TX

Experimental realization of crossover in shape and director field of nematic tactoids Jun 2015
ACS Colloid & Surface Science Symposium, Carnegie Mellon University, Pittsburgh, PA

TEACHING AND EDUCATIONAL CONTRIBUTIONS

Georgia Tech Courses, Instructor of Record

AI for Experimental Chemical Engineers (CHBE 4803/8803); 36 students Fall 2026
Fluid Mechanics (CHBE 3205); 53 students Spring 2026
Fluid Mechanics (CHBE 3205); 40 students Fall 2025
AI for Experimental Chemical Engineers (CHBE 4803/8803); 19 students Spring 2025
Fluid Mechanics (CHBE 3205); 21 students Fall 2024
Fluid Mechanics (CHBE 3205); 44 students Spring 2024
Transport Phenomena I (CHBE 3200); 44 students Fall 2023
Transport Phenomena I (CHBE 3200); 33 students Fall 2022

Earlier Teaching Experience, Rice University

Dean's Teaching Assistant, Thermodynamics I (CHBE 411) Fall 2014
Teaching Assistant, Colloidal and Interfacial Phenomena (CHBE 560) Spring 2014
Teaching Assistant, Transport Phenomena I (CHBE 401) Fall 2012
Teaching Assistant, Chemical Engineering Laboratory II (CHBE 433) Fall 2011

Course Development and Teaching Leadership

- Designed and introduced *AI for Experimental Chemical Engineers* in Spring 2025, a course associated with the Applications of AI and ML in Chemical Engineering minor and the Graduate Data Science Certificate
- Awarded a College of Engineering AI Makerspace grant for course development
- Undergraduate faculty advisor to sophomores, School of Chemical and Biomolecular Engineering, Georgia Tech (2023–present)

RESEARCH MENTORING AND STUDENT GUIDANCE

Ph.D. Research Advisees

Ziyi (Lucy) Liu, ChBE, Georgia Tech Fall 2026–present
Co-advised with Amirali Aghazadeh and Lily Cheung. Tentative thesis: *Liquid–Liquid Phase-Separated Peptide Coacervates for Sustainable Coatings*

Tianhua Zhao, ChBE, Georgia Tech — Fall 2026–present
Tentative thesis: *Multi-Agent AI for Planning and Control of Self-Driving Labs*

Yuling Zheng, Bioengineering, Georgia Tech — Fall 2025–present
Passed qualifying examination. Tentative thesis: *Cryogenic and Liquid-Phase Electron Microscopy Methods for Imaging Protein Macromolecules and Their Assemblies*

Isabel Panicker, ChBE, Georgia Tech — Fall 2023–present
Passed Ph.D. proposal. NSF GRFP honorable mention; Eckert Fellowship. Tentative thesis: *A Correlative Microscopy Platform for Studying Non-Equilibrium Dynamics of Active Matter*

Zain Shabeeb, ChBE, Georgia Tech — Fall 2022–present
Passed Ph.D. thesis proposal. GT LEAD Fellow; Eckert Fellowship. Tentative thesis: *AI-driven Electron Microscopy for Probing Nanoscale Biological Dynamics*

Undergraduate Research Trainees , Georgia Tech

Milan Barnes, ChBE	Fall 2026–present
Sardhak Boddu, ChBE	Fall 2026–present
Neha Muralitharan, ChBE	Fall 2026–present
Caroline Tzur, ChBE	Fall 2026–present
Benjamin Brough, ChBE	Fall 2026–present
Nathaniel (Nate) Glanzmann, ChBE	Fall 2026–present
Gabriela Briceno, Materials Science and Engineering	Summer–Fall 2026
Riya Patel, ChBE	Spring 2026–present
Siddarth Sreeram, Computer Engineering	Fall 2025–present
Diya Mahesh, ChBE	Spring 2025–present
Mahika Raj, Computer Science	Summer 2026
Anika Patel, Computer Science	Spring–Summer 2026
Alexander Wang, Computer Science	Summer 2025–Summer 2026
Max Xu, Computer Science	Summer 2025–Spring 2026
Madison Tat, ChBE; Presidential Undergraduate Research Award (PURA) fellow	Summer–Fall 2025
Risha Goel, ChBE; PURA fellow; now Ph.D. student at the University of Minnesota	Fall 2024–Spring 2025
Laurence Lines, ChBE	Fall 2024–Spring 2025
Isabella Faciano, ChBE	Fall 2024–Spring 2025
Lauren Caldwell, ChBE	Fall 2024–Spring 2026
Matthew Abernathy, ChBE	Fall 2024
Cecilia Reed, ChBE	2023–2025
Gabriel Joaquim Sampaio de Almeida, ChBE	Fall 2023
Arko Roy, ChBE	Fall 2023
Naisargi Goyal, ChBE; PURA fellow; now Ph.D. student at Princeton University	Fall 2022–Spring 2024

Earlier Research Trainees and Interns

Tanner Yamada, University of California, Berkeley, Chemical and Biomolecular Engineering	2018–2019

Evan Biggers, Rice University, Chemical and Biomolecular Engineering 2016–2017

Miranda Mittleman, high school summer intern, Rice University Summer 2017

Samuel Quitzau, NSF Research Experiences for Undergraduates (REU), Rice University 2016–2017

Graduate Thesis and Dissertation Committees

Ahmed Mohamed, Materials Science and Engineering; advisors: Natalie Stingelin and Juan Pablo Correa-Baena 2026–present

Seongeun Kim, ChBE; advisors: Lily Cheung and Shu Takayama 2026–present

Evan Moore, ChBE; advisor: Victor Breedveld 2025–present

Sayan Bhowmik, ChBE; advisor: AJ Medford 2024–present

Jiaji Lin, ChBE; advisor: Krista Walton; defended 2024–2026

Myeongsoo Kim, Bioengineering; advisor: Stanislav Emelianov; defended 2023–2025

Sushree Jagriti Sahoo, ChBE; advisor: AJ Medford; defended 2024

Sydney Wimberley, ChBE; advisor: Julie Champion; defended 2024–2025

Natechanok Yutthasaksunthorn, ChBE; advisor: Sankar Nair; defended 2023–2026

Zhitao Chen, ChBE; advisor: Nian Liu; defended 2023–2025

Ifayoyinsola A. Ibikunle, ChBE; advisor: David Sholl; defended 2023

PROFESSIONAL SERVICE, SCIENTIFIC LEADERSHIP, AND OUTREACH

Conference Organization and Session Leadership

Organizer and chair, DSOFT/DBIO Mini-Symposium, APS Global Physics Summit 2027

Session organizer and chair, AIChE Fall Meeting, Nanoscience and Engineering Forum 2026

Session organizer and chair, AIChE Fall Meeting, Nanoscience and Engineering Forum 2025

Session chair, ACS Colloid and Surface Science Symposium 2025

Session organizer and chair, AIChE Fall Meeting, Nanoscience and Engineering Forum 2024

Organizer, Atlanta Soft Matter Day Symposium 2024

Session co-chair, International Microscopy Congress 2023

Session co-chair, Microscopy and Microanalysis 2022

Session co-chair, AIChE Annual Meeting 2017

Editorial, Refereeing, and Grant Review Service

Guest Editor, *Journal of Visualized Experiments* (JoVE) 2022–present

Journal reviewer: *Science*; *Science Advances*; *Journal of the American Chemical Society*; *ACS Nano*; *Soft Matter*; *Physical Review E*; *PRX Life*; *ACS Applied Electronic Materials*; *ACS Applied Polymer Materials*; *Journal of Rheology*; *Physical Chemistry Chemical Physics*; *Nanoscale Horizons*; *Digital Discovery*; *Advanced Intelligent Systems*

Conference program committee/reviewer: NeurIPS AI4Mat; ICML AI4Mat

Panelist, NSF CAREER, CBET program 2024

Reviewer, Graduate Women in Science Fellowship 2021

Georgia Tech Committee Service

Information Technology Committee, School of ChBE 2026–present

Faculty Search Committee, School of ChBE 2025–present

ChBE Advisory Committee, School of ChBE 2024–2026

Graduate Admissions Committee, School of ChBE 2024–2025

Facilities Advisory Committee, Institute for Matter and Systems 2024–present

Seminar Organization Committee, School of ChBE 2022–2025

Graduate Recruitment Committee, School of ChBE 2022–2024

Sophomore undergraduate faculty advisor, School of ChBE 2023–present

Bioengineering Day Thesis Competition Judge 2023

Research and Academic Program Development

Co-lead, AI for Chemicals and Materials Discovery Initiative, Institute for Data Engineering and Science, Georgia Tech 2023–2025

Research program lead, Compressed Super-Resolution TEM Using a Nanoelectronic Coded Aperture Device, Georgia Tech 2024–present

Program faculty: Machine Learning Ph.D. Program and Bioengineering Ph.D. Program, Georgia Tech

Institute affiliations: Institute for Matter and Systems; Institute for Bioengineering and Biosciences; Institute for Data Engineering and Science; Strategic Energy Institute

Professional Society Memberships

- American Institute of Chemical Engineers (AIChE); American Chemical Society (ACS); American Physical Society (APS); Materials Research Society (MRS); Microscopy Society of America (MSA)

Outreach and Public Engagement

Hosted K–12 teacher through NNCI Research Experience for Teachers (RET) program, Jamali Group 2026

DyeNamic Lab, Atlanta Science Festival, Atlanta, GA 2025–2026

Hosted tours of the Georgia Tech electron microscopy facility for Campbell and Walton High School students

Panelist, UC Berkeley Visiting Scholar and Postdoctoral Association, Academic Job Search Strategies Workshop 2024

Panelist, NextProf program 2023

Panelist, Georgia Tech AIChE Women in Chemical Engineering 2023

STEM GEM workshop panelist/presenter, Atlanta, GA 2023

Postdoctoral mentorship and career guidance sessions, Gordon Research Conference on Liquid-Phase Electron Microscopy 2022

Postdoctoral representative, Chemistry Graduate Life Committee, UC Berkeley 2019–2020

Recitation chair, Chemical and Biomolecular Engineering Graduate Student Association, Rice University 2013–2014

Workshop presenter, Sally Ride Science Festival for Girls, Houston, TX 2012, 2013

SELECTED RESEARCH COVERAGE AND PRESS

Georgia Tech Research News: Study Uncovers New Insights into Nanoparticle Motion. Jun. 5, 2026
Also featured on the Georgia Tech Institute for Matter and Systems news site

Phys.org (syndicated from Georgia Tech): Agentic AI Could Help Electron Microscopes Plan, Adapt, and Analyze Experiments. May 22, 2026
Further online aggregation: Science Feed (AI-generated summary)

Georgia Tech Research News: Accelerating Discovery with AI. May 18, 2026
Also republished by the Georgia Tech College of Sciences

Medical Xpress: Nanohydrogels Steer Cancer Drugs to Tumors, Aiming to Spare Healthy Tissue Mar. 10, 2026

Georgia Tech: Target the Tumor. Spare the Body Mar. 6, 2026

Phys.org: New AI Tool Deciphers Mysteries of Nanoparticle Motion in Liquid Environments July 15, 2025

Georgia Tech: New AI Tool Deciphers Mysteries of Nanoparticle Motion in Liquid Environments July 11, 2025

Rice University: Molecular Jiggling Has Implications for Carbon Nanotube Fibers May 31, 2021

Phys.org: Molecular Jiggling Has Implications for Carbon Nanotube Fibers May 31, 2021

Materials Today: Scientists Explore Ways to Produce High-Quality Fibers from Carbon Nanotubes Nov. 10, 2017

Phys.org: Researchers Advance Characterization, Purification of Nanotube Wires and Films Oct. 16, 2017

EurekAlert!: Long Nanotubes Make Strong Fibers Oct. 16, 2017

PROFESSIONAL REFERENCES

Available upon request